

Executive Summary: Project Lighthouse Reconciliation

To: Chief Financial Officer & Senior Leadership

From: Vishal Patil, Data Analytics

Subject: Post-Submission Review of HMDA LAR Reporting Pipeline

1. The Bottom Line: What Actually Happened?

A post-submission reconciliation of the HMDA LAR data revealed a **systemic pipeline synchronization failure**, rather than a localized human underwriting error or actual shifting market trajectory.

- **Population Sync Issues:** The Submitted file was cut at the wrong time in the data replication cycle. As a result, 9,929 "unaccounted" applications (7.6% of the volume) were incorrectly included in the regulatory submission.
- **Mapping Issues:** For the valid loans that *did* exist in both files, the data engineering logic pulled from the wrong database tables. This caused 1 in 5 loans to report incorrect values for structural underwriting fields like Combined Loan-To-Value (CLTV) and Loan Term.

2. What Materially Changes the Story?

Surprisingly, nothing. The noise is loud, but the strategic impact is minimal.

While the volume of data errors is unacceptably high for regulatory standards (and requires immediate QC architecture fixes), the narrative that the company's performance or risk profile radically shifted is *definitively false*.

- **The Borrower Profile Didn't Change:** Median Income stayed exactly at \$100.00k across both files.
- **The Outcomes Didn't Shift:** Approval Rates for matched records remained perfectly flat at **76.40%**. The **47.8 bps** drop in the overall portfolio was purely a mathematical artifact caused by purging high-approval discarded populations.
- **The Pricing Signal Moved Appropriately:** The only true directional shift was in CLTV, rising from 77.0 up to 80.0—a logical standardization toward the industry-standard PMI cutoff. The CFO does not need to change strategy; they just need to change the data pipeline.

3. Signal vs. Noise (What to Ignore vs. What to Audit)

- **The Noise (Ignore):** The population drop of 9,929 records. These discarded records were disproportionately high-income (\$104k) and high-approval (82.9%), which mechanistically explains the entire portfolio drop without requiring 'random noise'.
- **The Signal (Follow-Up Required):** An outcome inversion in the "Home Improvement" segment. While total company approvals dropped by 47.8 bps, Home Improvement approvals inexplicably *increased* by 49.3 bps. This pinpoints exactly where the analytical mapping logic is fracturing.

4. Scalable Validation & QC Pipeline Architecture

To prevent submitting unverified data in the future, we must shift from reactive audits to proactive pipeline constraints. If this analysis ran quarterly, I would deploy the following architecture:

A. Where Controls Sit in the Data Pipeline:

Regulatory reporting should never pull from a volatile "live" system. We must establish an immutable **"Regulatory Data Mart."** Once a loan reaches a terminal state (Funded/Denied), its data is structurally locked.

B. Automated vs. Manual Checks:

- **Automated (Continuous):** Implement data contract tools like dbt directly on the SQL warehouse. We standardise automated generic tests: `not_null` assertions on critical fields and threshold boundaries (e.g., CLTV never physically exceeding 200%).
- **Manual (Human Gatekeeper):** Compliance sign-off must remain the only manual step. If an automated drift alert fires, a human must document the SQL root cause before the pipeline unlocks the data for transmission to the CFPB.

C. Outputs Stakeholders Receive:

- **Data Engineering:** Real-time Slack/Email alerts if a pipeline merge triggers a >1% historical drift.
- **Compliance/CFO:** A monthly automated Tableau dashboard showing a "Traffic Light" status (Red/Yellow/Green) for the top 5 regulatory fields, eliminating end-of-year surprises.

5. Final Recommendation & Regulatory Reliance

If presenting this to Internal Audit, I would request two immediate follow-ups before closing the book:

1. **Source System Tie-Out:** Pull the raw Loan Origination System (LOS) extract for a random sample of 100 heavily altered rows to definitively prove the Revised file logic aligns structurally with the LOS system of record.
2. **Geographic Mapping Audit:** Investigate why the data standardisation incorrectly inflated California income by ~\$1.2k while suppressing East Coast income by ~\$1.5k.

Based on these findings, would I rely on this dataset for regulatory or executive reporting?

No—not without completing the Source System Tie-Out.

The risk is highly asymmetric. Submitting an incorrect dataset to the CFPB over high-scrutiny pricing fields exposes the institution to severe civil money penalties and reputational damage. While the *Revised* file is vastly superior to the Submitted file, it cannot be legally blessed for external reporting until the Data Engineering team proves their new SQL logic maps 1:1 against the raw Loan Origination System.